

Computer Organization and Architecture		
Lab #:	1	
Lab Title:	Marks of 10 students and then displays the average. 	

Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

Lab : 01

➤ Task #1

Write a program that takes marks of 10 students and then displays the average.

Code:

```
.data
marks: .space 40    # Reserve space for 10 integers (4 bytes each)

.text
.globl main

main:
    # Initialize the loop counter and sum
    li $t0, 0        # $t0 will be the loop counter (i)
    li $t1, 0        # $t1 will hold the sum of the marks

input_loop:
    # Prompt for input and store in the array
    li $v0, 5        # syscall read_int
    syscall          # read an integer
    sw $v0, marks($t0) # store the integer at marks[i]
    # Add the input value to the sum

    add $t1, $t1, $v0
    # Increment the loop counter and check if we have input 10 marks
    addi $t0, $t0, 4    # move to the next integer in the array (4 bytes each)
    li $t2, 40         # 10 * 4 = 40 (10 integers)
    bne $t0, $t2, input_loop # if i != 10, continue the loop

    # Calculate the average
    li $t3, 10         # load the number of students into $t3
    div $t1, $t3        # divide the sum by 10
    mflo $t4           # move the quotient (average) to $t4

    # Print the average
    move $a0, $t4       # move the average to $a0 for printing
    li $v0, 1          # syscall print_int
    syscall            # print the average

    # Exit the program
    li $v0, 10         # syscall code for exit
    syscall            # exit
```

➤ Task #2

Write a program that inputs three 1-digit numbers and combines them to form a 3 digit number. For example input 4, 8 ,7, output=487.

Code:

```
.data
.text
.globl main

main:
    # Read the first digit
    li $v0, 5          # syscall read_int
    syscall            # read the first digit
    move $t0, $v0       # store the first digit in $t0

    # Read the second digit
    li $v0, 5          # syscall read_int
    syscall            # read the second digit
    move $t1, $v0       # store the second digit in $t1

    # Read the third digit
    li $v0, 5          # syscall read_int
    syscall            # read the third digit
    move $t2, $v0       # store the third digit in $t2

    # Combine the digits to form the 3-digit number
    li $t3, 100         # Load 100 into $t3 (for multiplying the first digit)
    mul $t0, $t0, $t3    # Multiply the first digit by 100
    li $t3, 10          # Load 10 into $t3 (for multiplying the second digit)
    mul $t1, $t1, $t3    # Multiply the second digit by 10
    add $t4, $t0, $t1    # Add the result of the first two digits
    add $t4, $t4, $t2    # Add the third digit to form the final number

    # Print the 3-digit number
    move $a0, $t4        # Move the result to $a0 for printing
    li $v0, 1           # syscall print_int
    syscall              # print the number

    # Exit the program
    li $v0, 10          # syscall to exit
    syscall              # exit
```

➤ Task #3

Write a program that inputs marks of four quizzes and four assignments, each out of 10, and calculates total marks out of 25. Total quiz marks 15/25 and Assignment marks = 10/25

```
.data
quiz_weight: .float 0.6    # Quiz weight = 15/25 = 0.6
assignment_weight: .float 0.4 # Assignment weight = 10/25 = 0.4

.text
.globl main

main:
    # Initialize the loop counters and total sums
    li $t0, 0        # $t0 will be the loop counter for quizzes
    li $t1, 0        # $t1 will be the loop counter for assignments
    li $t2, 0        # $t2 will hold the sum of the quiz marks
    li $t3, 0        # $t3 will hold the sum of the assignment marks

    # Input quiz marks
input_quiz_loop:
    li $v0, 5        # syscall read_int
    syscall          # read the quiz mark
    add $t2, $t2, $v0 # add the quiz mark to the total

    addi $t0, $t0, 1 # increment the quiz counter
    li $t4, 4        # total 4 quizzes
    bne $t0, $t4, input_quiz_loop # continue loop if less than 4 quizzes

    # Input assignment marks
input_assignment_loop:
    li $v0, 5        # syscall read_int
    syscall          # read the assignment mark
```

```

add $t3, $t3, $v0 # add the assignment mark to the total

addi $t1, $t1, 1  # increment the assignment counter
li $t5, 4          # total 4 assignments   bne $t1, $t5, input_assignment_loop # continue
loop if less than 4 assignments

# Calculate weighted quiz marks (total out of 15)
li $v0, 2          # syscall to load a floating-point constant
la $a0, quiz_weight # load the quiz weight (0.6) address into $a0
lwc1 $f12, 0($a0)  # load quiz weight into $f12
mtc1 $t2, $f14     # move quiz sum from $t2 to $f14 (as integer)
cvt.s.w $f14, $f14 # convert integer in $f14 to floating-point
mul.s $f14, $f14, $f12 # multiply quiz sum by the quiz weight
cvt.w.s $f14, $f14 # convert result back to integer
mfc1 $t6, $f14     # move the result back to integer register $t6

# Calculate weighted assignment marks ( out of 10)
li $v0, 2          # syscall to load a floating-point constant
la $a0, assignment_weight # load the assignment weight (0.4) address into $a0
lwc1 $f12, 0($a0)  # load assignment weight into $f12
mtc1 $t3, $f14     # move assignment sum from $t3 to $f14 (as integer)
cvt.s.w $f14, $f14 # convert integer in $f14 to floating-point
mul.s $f14, $f14, $f12 # multiply assignment sum by the assignment weight
cvt.w.s $f14, $f14 # convert result back to integer
mfc1 $t7, $f14     # move the result back to integer register $t7

# Calculate total marks out of 25
add $t8, $t6, $t7  # add weighted quiz and assignment marks

# Print the total marks
move $a0, $t8      # move the total marks to $a0 for printing
li $v0, 1          # syscall print_int
syscall            # print the total marks

```

Exit the program

li \$v0, 10 # syscall to exit

syscall # exit